$(a+1)^2 = a^2 + 2a + 1$

Now that's a polynomial. That does not look very special, does it? We have seen a lots of algebraic expressions already, so why to bother about these? There are many reasons why polynomials are interesting and important in mathematics.

For now, we will just take one example showing their use. Remember, we studied lots of arithmetic and then came to algebra, thinking of variables as unknown numbers. Actually we can now get back to numbers and try to write them in the language of algebra.

Consider a number like 5418. It is actually 5 thousand 4 hundred and eighteen. Write it as:

$5 \times 1000 + 4 \times 100 + 1 \times 10 + 8$

which again can be written as:

$5 \times 10^3 + 4 \times 10^2 + 1 \times 10^1 + 8$

Now it should be clear what this is about. This is of the form $5x^3 + 4x^2 + x + 8$, which is a polynomial. How does writing in this form help? We always write numbers in decimal system, and hence always $x = 10$. Then what is the fun? Remember divisibility rules? Recall that a number is divisible by 3 only if the sum of its digits is divisible by 3. Now notice that if x divided by 3 gives 1 as remainder, then it is the same for x^2, x^3, etc. They all give remainder 1 when divided by 3. So you get each digit multiplied by 1, added together, which is the sum of digits. If that is divisible by 3, so is the whole number. You can check that the rule for divisibility by 9, or even divisibility by 2 or 5, can be proved similarly with great ease. Our purpose is not to prove divisibility rules but to show that representing numbers as polynomials shows us many new number patterns. In fact, many objects of study, not just numbers, can be represented as polynomials and then we can learn many things about them.

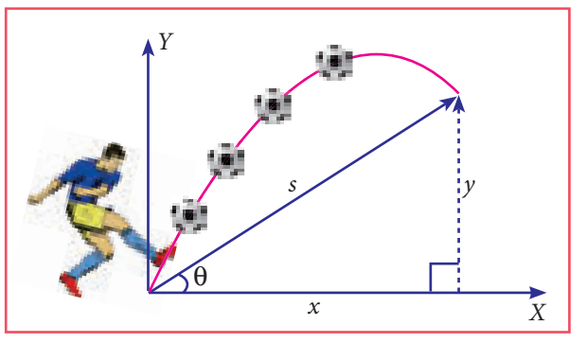


Fig. 3.1

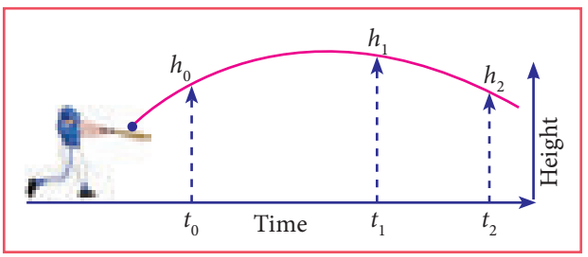


Fig. 3.2

In algebra we think of x^2, $5x^2-3$, $2x+7$ etc as *functions* of x. We draw pictures to see how the function varies as x varies, and this is very helpful to *understand* the function. And now, it turns out that a good number of functions that we encounter in science, engineering, business studies, economics, and of course in mathematics, all can be approximated by polynomials, if not actually be represented as polynomials. In fact, approximating functions using polynomials is a fundamental theme in all of higher mathematics and a large number of people make a living, simply by working on this idea.

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